rm Command in Linux

rm command `rm` Command in Linux

The `rm` command is used to **remove (delete) files and directories** in Linux.
[WARNING] Unlike GUI-based systems, files deleted with `rm` are **not moved to Trash/Recycle Bin**. Once removed, they are gone permanently.

- Syntax

rm [options] file_name

- `file_name` -> File or directory you want to delete
- `options` -> Modify how the deletion works (interactive, recursive, force, verbose, etc.)

- Options with Examples

1. Remove a File

rm file.txt

[NOTE] Deletes `file.txt` from the current directory.
-> If the file does not exist, you'll get an error.

2. Remove Multiple Files

rm file1.txt file2.txt file3.txt

[NOTE] Deletes `file1.txt`, `file2.txt`, and `file3.txt` at once.

3. Remove Files with a Pattern (Wildcard `*`)

rm *.log

[NOTE] Deletes **all files ending with `.log`** in the current directory.

-> Useful for cleaning log files.

4. Interactive Delete (`-i`)

rm -i file.txt

[NOTE] Asks for confirmation before deleting `file.txt`.

rm: remove regular file 'file.txt'? y

-> Good for safety when you don't want to delete files accidentally.

5. Verbose Delete (`-v`)

rm -v file.txt

[NOTE] Shows a message when deleting files.

Output:

removed 'file.txt'

6. Force Delete (`-f`)

rm -f file.txt

[NOTE] Deletes `file.txt` **without asking** and **without showing errors** if the file doesn't exist.

-> Useful in scripts to avoid interruption.

7. Recursive Delete (`-r` or `-R`)

rm -r my_folder/

[NOTE] Deletes the directory `my_folder` **and all its files/subdirectories**.

-> Without `-r`, `rm` cannot delete directories.

8. Recursive + Force (`-rf`)

rm -rf my_folder/

[NOTE] Deletes `my_folder` and all contents **forcefully and without confirmation**.

-> Most commonly used, but [WARNING] **very dangerous**.

9. Recursive + Verbose (`-rv`)

rm -rv my_folder/

[NOTE] Deletes the folder and prints the name of each file being removed.

Output:

removed 'my_folder/file1.txt'
removed directory 'my_folder'

10. Delete Multiple Directories

rm -rf folder1/ folder2/

[NOTE] Deletes both `folder1` and `folder2` with all their contents.

[WARNING] Dangerous Example (Never Run!)

rm -rf /

[NOTE] This will attempt to delete **everything from the root directory**, making your system unusable.

- Pro Safety Tip

You can make `rm` safer by adding an **alias** in your `.bashrc` or `.zshrc`:

alias rm='rm -i'

[NOTE] This makes `rm` always **ask before deleting**.
-> To bypass, you can still run `\rm file.txt`.

[CHECK] **Summary:**
- Use `rm file` -> delete a file
- Use `rm -i` -> safe delete (asks confirmation)
- Use `rm -r` -> delete a directory
- Use `rm -rf` -> delete directory forcefully ([WARNING] careful!)
- Use `rm -v` -> show what's being deleted